

BPM2_ESOFT QUIZ 2_S2026_GD

of Questions: 30

Question #: 1

A 56-year-old man comes to the physician because of a 3-month history of nausea, weight loss and abnormal bowel movements. Physical examination shows an emaciated patient, with abdominal distension and a palpable mass in the right upper quadrant. CT scan of the abdomen shows a tumor in the right hepatic flexure of the colon. Based on the location of the tumor, which of the following groups of lymph nodes will most likely receive metastasizing cells from the cancerous lesion?

- A. Internal iliac
- B. Inferior mesenteric
- C. Celiac
- ✓D. Superior mesenteric
- E. Superficial inguinal

Rationale: The right hepatic flexure is a derivative of the midgut. It will be drained by the midgut vein, which is the superior mesenteric vein. Therefore the lymphatic drainage will be to the superior mesenteric lymph nodes, since lymphatic drainage follows the venous drainage.

Question #: 2

A 30-year-old woman comes to the physician for a routine examination. She is in her third trimester. Ultrasound of the fetus shows herniation of part of the gut tube into the left hemithorax. A diagnosis of congenital diaphragmatic hernia is made. Which of the following is most likely caused by this condition?

- A. Respiratory distress syndrome
- B. Hemothorax
- C. Pneumothorax
- ✓D. Pulmonary hypoplasia
- E. Eventration of the diaphragm

Rationale: Failure of development of the pleuripерitoneal membranes leads to congenital diaphragmatic hernia. The abdominal contents herniate into the thoracic cavity, which takes up

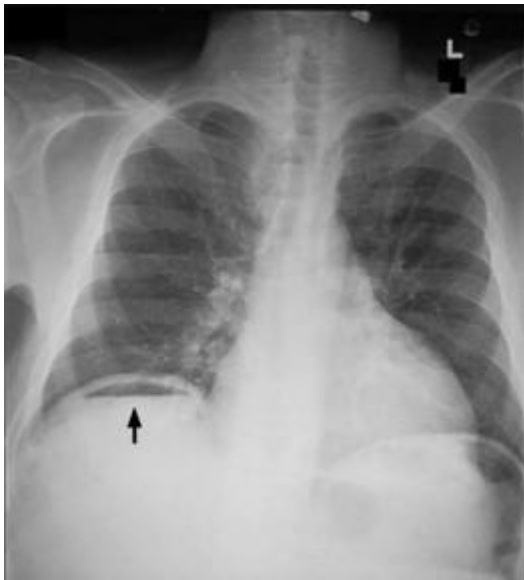
the space within this cavity and compresses the contents of this cavity. This results in there being limited space for development of the lungs which leads to pulmonary hypoplasia.

Question #: 3

A 25-year-old man is brought to the emergency department because of a penetrating stab wound to the abdomen that he received 30 minutes ago in a brawl. Physical examination shows a large shard of wood penetrating the abdominal wall. A radiograph of the abdomen is attached. Which of the following organs is most likely punctured resulting in the abnormality indicated by the arrow in the image?

- A. Liver
- ✓B. Stomach
- C. Spleen
- D. Pancreas

Rationale: Pneumoperitoneum is the presence of air or gas in the abdominal (peritoneal) cavity. A gastrointestinal perforation constitutes one of the commonest causes.



Question #: 4

A 2-day-old boy is brought the emergency department by his mother because of severe vomiting after feeding for past 12 hours. Endoscopic studies shows an

external constriction at the second part of the duodenum. Which of the following is the most likely diagnosis?

- A. Duodenal atresia
- ✓B. Annular pancreas
- C. Duodenal stenosis
- D. Pyloric stenosis

Rationale: This is a ring like part of the pancreas that surrounds the descending (2nd part) of the duodenum. It results from growth of a bifid ventral pancreatic bud around the duodenum. The parts of the bifid ventral bud fuse with the dorsal bud, forming a pancreatic ring. This can result in complete or partial bowel obstruction.

Question #: 5

A 58-year-old man comes to the physician because of a 2-week history of difficulty swallowing. Radiographic studies show an enlarged left atrium impinging on structures in the thorax, especially the esophagus and esophageal plexus. Which of the following structures most likely contains the preganglionic sympathetic fibers that provide innervation to the affected organ?

- A. Vagus nerve
- B. Lateral horn like structures in S2-S4
- ✓C. Lateral horn T1-T4
- D. Paravertebral ganglia T1-T4
- E. Pre aortic ganglion

Rationale: Preganglionic sympathetic cell bodies are found in the lateral horn from T1 to L2. Specifically for the esophagus, the fibers come from the cardiopulmonary plexus, which receives sympathetics from T1 to T4.

Question #: 6

An 80-year-old man comes to the emergency department because of a 2-hour history of severe abdominal pain. A CT scan of the abdomen shows ischemic bowel. During laparotomy, a region of ischemic bowel is seen extending from the left colic flexure of the transverse colon to the beginning of the sigmoid colon. Which of the following nervous structures most likely carries the visceral afferent fibers responsible for the patient's pain to the CNS?

- A. Pelvic splanchnics
- ✓B. Lumbar splanchnics
- C. Vagus
- D. Lesser splanchnics
- E. Greater splanchnics

Rationale: The sympathetic innervation to the area of left colic flexure to the sigmoid colon (hindgut) is from L1- L2, the lumbar splanchnic nerves. Visceral afferent fibers from this part of the hindgut are carried back to the spinal cord by these sympathetic fibers.

Question #: 7

A 21-year-old woman comes to the emergency department because of a 2-hour history of headache and dizziness. She is at the 17th hour mark of an 18:6 intermittent fasting schedule. She has not had fever, chills, or shortness of breath. She has no personal or family history of major medical illnesses. She has never smoked cigarettes. Physical examination shows no abnormalities. Laboratory studies show plasma glucose level is 72 mg/dL (70-100 mg/dL). Which of the following is most likely observed in her liver during her current state?

- A. Inhibition of AMP-kinase by glucagon
- B. Allosteric inhibition of fructose 1,6-bisphosphatase (FBPase-1) activity
- ✓C. Allosteric activation of pyruvate carboxylase by acetyl-CoA
- D. Covalent modification and inhibition of glycogen phosphorylase
- E. High NADH concentration, which activates pyruvate dehydrogenase
- F. Elevated cytosolic citrate concentration and activation of acetyl-CoA carboxylase

Rationale: The patient is fasting and would have a low insulin: glucagon ratio.

The pathways that would be active in the liver in the fasting state: Glycogenolysis and gluconeogenesis (which maintain plasma glucose concentration). Beta-oxidation of fatty acids and ketogenesis would also be active.

Glycolysis, glycogen synthesis, and fatty acid synthesis would be less active.

AMP-kinase would be activated by glucagon. AMP-kinase phosphorylates acetyl-CoA carboxylase and inhibits it.

Low cytosolic citrate during fasting promotes the formation of the monomer of acetyl-CoA carboxylase. Both of these inhibit fatty acid synthesis.

Fructose 1,6-bisphosphatase is active due to a low concentration of fructose 2,6-bisphosphate.

The bifunctional enzyme is phosphorylated, and PFK-2 is less active in the fasting state.

Acetyl-CoA, formed by fatty acid oxidation (beta-oxidation), allosterically activates pyruvate carboxylase (key enzyme of gluconeogenesis).

Glycogen phosphorylase is active in the phosphorylated state during fasting. Note that liver glycogen stores are depleted by 17-24 hours of fasting, and gluconeogenesis is the sole pathway responsible for the maintenance of the plasma glucose concentration.

Pyruvate dehydrogenase is less active due to NADH provided by the beta-oxidation of fatty acids.

Question #: 8

An 18-month-old boy is brought to a physician by his parents because of a 6-month history of failure to thrive and tiring quickly when playing. He has had two previous episodes of seizures. He is at the 5th percentile for height and weight. Physical examination shows a protuberant abdomen with a palpable liver margin 2 cm below the right costal margin. Laboratory studies show hyperuricemia, lactic acidemia, elevated serum triglycerides (TAG), and hypoglycemia. Which of the following enzyme deficiency disorders is most likely in this patient?

- A. Phosphofructokinase-1
- B. Glycogen branching enzyme
- C. Glycogen debranching enzyme
- ✓D. Glucose 6-phosphatase
- E. Glucokinase
- F. Alpha 1,6-glucosidase

Rationale: The patient most likely has von Gierke disease due to a deficiency of glucose 6-phosphatase. Fasting hypoglycemia, hyperuricemia, elevated serum lactate concentration, and hypertriglyceridemia are characteristic findings in these patients. Liver biopsy shows accumulation of glycogen with normal branching pattern. Gluconeogenesis and glycogenolysis are affected as glucose 6-phosphatase participates in both these processes.

Patients with PFK-1 deficiency (Tarui disease) have no hypoglycemia. They have pain during anaerobic exercise and hemolysis. Similar to McArdle disease.

Patients with glycogen branching enzyme deficiency (Andersen disease) have liver failure and generally no hypoglycemia. Glycogen with an abnormal structure (long and unbranched) accumulates.

Patients with Cori disease (Glycogen debranching enzyme deficiency) may have mild hypoglycemia and muscle weakness. The other lab findings are generally not present. Glycogen

with an abnormal structure (short branches; limit dextrin) accumulates in the liver and muscle. Glucokinase deficiency results in hyperglycemia (Monogenic diabetes; MODY-2) and not hypoglycemia.

Pompe disease is a lysosomal glycogen storage disorder due to deficiency of lysosomal 1-6 glucosidase. Characteristic features are hypotonia, cardiomegaly, and feeding difficulties. Patients do NOT have hypoglycemia.

Question #: 9

A 10-month-old boy is brought to the emergency department by his parents because of a sudden-onset of seizures. During the past 2 days, he has had fever and cough. Results of laboratory studies obtained on admission are shown in the table. Which of the following disorders is most likely in this patient?

- A. Refsum disease
- B. Zellweger syndrome
- ✓C. Medium chain acyl CoA dehydrogenase (MCAD) deficiency
- D. Myopathic carnitine deficiency
- E. Carnitine palmitoyl transferase - II (CPT-II) deficiency
- F. Glucagon secreting tumor

Rationale: The vignette describes a patient with MCAD deficiency (systemic beta oxidation defect. Note the severe hypoglycemia and low ketone body levels (hypoglycemia and hypoketosis). The patient is in metabolic crisis which generally follows a period of stress/ illness or starvation. In a patient with MCAD deficiency, there is impaired beta-oxidation in the liver. As a result there is less acetyl CoA formation. Acetyl CoA is an absolute activator of pyruvate carboxylase (Gluconeogenesis). Less [acetyl CoA] impairs activation of pyruvate carboxylase and impairs gluconeogenesis resulting in the severe hypoglycemia. Impaired glucose supply to the brain due to hypoglycemia can manifest as lethargy or seizures (neuroglycopenic effect). Less [acetyl CoA] provides reduced substrate for ketogenesis in the liver and hence the ketone body production is also impaired and ketone body levels are low in these patients. Other laboratory findings are elevated levels of C8-C10 acyl carnitine in blood and urine. Dicarboxylic acids are also elevated in blood due to increased flux of fatty acids via omega-oxidation. Refsum disease is due to a defect in peroxisomal alpha oxidation. It is characterized by elevated levels of phytanic acid (branched chain fatty acids) in blood. Zellweger syndrome is due to a defect in peroxisome formation. All peroxisome functions are impaired. Characterized by elevated levels of C-22 and C-24 (very-long chain) fatty acids in blood and urine. Myopathic beta-oxidation defects (myopathic carnitine deficiency and CPT-II deficiency) present in adolescents as muscle cramps during exercise. They are unable to tolerate sustained aerobic activity. Muscle biopsy shows TAG droplets.

Glucagon secreting tumors present with hyperglycemia and NOT hypoglycemia.

	Patient values	Reference ranges
Blood glucose	45	70-100 mg/dL
3-hydroxybutyrate	0.3	0.4-0.5 mmol/L

Question #: 10

A 14-year-old boy is brought to the physician because of a 3-week history of muscle cramps. Examination of a muscle biopsy specimen shows normal glycogen levels but increased triglyceride (TAG) droplets. His plasma glucose and ketone body levels are within normal limits. Which of the following processes is most likely defective in this patient?

- A. Oxidation of very long chain fatty acids
- B. Oxidation of branched chain fatty acids
- C. Muscle glycogenolysis
- ✓D. Transport of fatty acids into mitochondria
- E. Endothelial lipoprotein lipase activity
- F. Adipose tissue lipolysis

Rationale: The vignette describes a patient with myopathic beta-oxidation defect. Note the history of muscle cramps on exercise. Presence of TAG droplets on muscle biopsy shows a defect in utilization of fatty acids for energy during exercise. Note additional findings in this patient would be increased lactate levels following strenuous exercise (Note the forearm ischemic test has a normal response; or increased lactate levels). This indicates that the glycogen degradation and anaerobic glycolysis and formation of lactate in muscle is not affected. Normal plasma glucose levels and normal ketone body levels indicate that the beta-oxidation defect is not involving the liver (not systemic). The patient most likely has a Carnitine palmitoyl transferase-II (CPT-II) deficiency or myopathic carnitine deficiency. These patients are unable to perform sustained aerobic exercise. Remember that energy for aerobic activity is generally provided by beta-oxidation of fatty acids.

Zellweger syndrome is characterized by defect in peroxisome formation. A characteristic finding is a defect in peroxisomal oxidation of very long chain fatty acids (C22-C24). All peroxisomal functions are impaired and it is a severe disorder that presents in infancy and is usually fatal. Liver and brain is usually affected.

Oxidation of branched chain fatty acids occurs in the peroxisomes by alpha-oxidation. This process is defective in Refsum disease. Characterized by elevated levels of branched chain

fatty acids (phytanic acid) in circulation. Presents in adolescents as blindness and ataxia. A defect in muscle glycogenolysis (McArdle disease) is characterized by muscle cramps during exercise. They are not able to perform anaerobic exercise. Blood lactate levels will NOT increase following anaerobic activity. Muscle biopsy shows accumulation of glycogen in muscle. This is an important differential diagnosis.

Endothelial lipoprotein lipase deficiency presents with elevated levels of chylomicrons and VLDL in circulation. Laboratory studies would show elevated fasting serum triacylglycerol (TAG). Increases risk of pancreatitis.

Hormone sensitive lipase deficiency is characterized by inadequate adipose tissue lipolysis. Patients would have low circulating free fatty acid levels in the fasting state.

Question #: 11

A 4-week-old boy is brought to the physician by his mother because of a 5-day history of vomiting and failure to thrive. He is exclusively breast-fed. The mother says that vomiting usually occurred within an hour of breast feeding. Physical examination shows jaundice and an enlarged liver with a palpable lower border of the liver. Examination shows a delay in the developmental milestones. The plasma glucose level is 50 mg/dL. Urinalysis shows the presence of a reducing sugar, which is not glucose. Laboratory studies show an inherited enzyme deficiency. Which of the following reactions is most likely impaired in this patient?

- A. Hydrolysis of lactose to glucose and galactose
- B. Conversion of fructose to fructose 6-phosphate
- C. Phosphorylation of galactose
- ✓D. Transfer of uridyl group to galactose 1-phosphate to form UDP-galactose
- E. Conversion of fructose 1-phosphate to glyceraldehyde and dihydroxyacetone phosphate
- F. Oxidation of branched chain fatty acids

Rationale: The vignette describes a child with classical galactosemia due to GALT deficiency. Note the hypoglycemia on breast feeding, involvement of the liver (jaundice and hepatomegaly) and neurological developmental delay. Presence of a reducing sugar in the urine is an indication that it is most likely galactose in urine.

GALT catalyzes the conversion of galactose 1-phosphate to UDP galactose (Transfer of UDP group).

Lactase deficiency (intolerance) is characterized by diarrhea. There is no reducing sugar in urine; no liver involvement and no delay in the developmental milestones.

Fructokinase catalyzes the conversion of fructose to fructose 1-phosphate. Deficiency results in benign fructosuria. Patients have reducing sugar in urine after sucrose ingestion but no other manifestations.

Aldolase B catalyzes the breakdown of fructose 1-phosphate to glyceraldehyde and

dihydroxyacetone phosphate. Aldolase B deficiency is characterized by severe hypoglycemia following sucrose intake. Reducing sugar (fructose) is present in urine. Untreated children have liver cell failure and jaundice.

Galactokinase deficiency results in non-classical galactosemia. The enzyme catalyzes the phosphorylation of galactose to form galactose 1-phosphate. Deficiency of galactokinase results in cataracts; NO liver involvement and NO developmental delay.

Oxidation of branched chain fatty acids is defective in Refsum disease. Characterized by a defect in alpha-oxidation of phytanic acid in peroxisomes. Laboratory studies show increased phytanic acid levels in blood.

Question #: 12

A 3-month-old girl is brought to the physician by her mother because of failure to thrive and feeding difficulties since birth. Physical examination shows macroglossia, hepatomegaly and congestive heart failure due to hypertrophic cardiomyopathy. Laboratory studies show elevated serum troponin, CK-MB, ALT and AST, and normal plasma glucose levels. Examination of a muscle biopsy specimen shows the presence of vacuoles that stain positive for a glucose polymer. Which of the following enzyme deficiency disorders is most likely in this patient?

- ✓A. Lysosomal alpha-1, 4-glucosidase
- B. Glycogen phosphorylase
- C. Glucose 6-phosphatase
- D. Glycogen debranching enzyme
- E. Phosphofructokinase-1

Rationale: The vignette describes a child with Pompe disease. Note the involvement of the heart and the accumulation of glycogen in the lysosomes. The enzyme deficient in this disorder is acid maltase also known as alpha 1,4 glucosidase.

Glucose 6-phosphatase deficiency presents as hypoglycemia, hepatomegaly (glycogen deposits in liver with normal branching pattern) and lactic acidosis.

Liver glycogen phosphorylase deficiency (Hers disease) presents as hypoglycemia and hepatomegaly (less severe than von Gierke)

Muscle glycogen phosphorylase deficiency presents as cramps during strenuous (anaerobic exercise); Glycogen deposits in muscle (normal branching pattern)

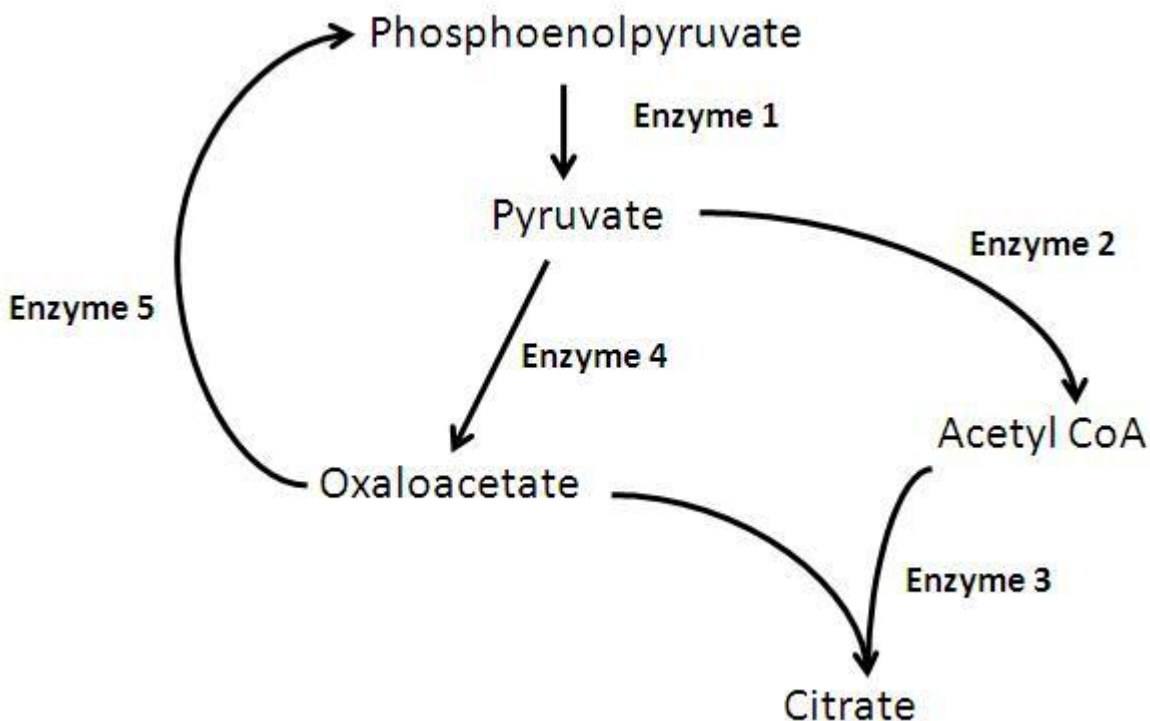
Glycogen debranching enzyme deficiency (Cori disease) - presents with hypoglycemia; Muscle weakness; Glycogen accumulation in liver and muscle with short branches (limit dextrinosis)

Question #: 13

An investigator studying liver metabolism in experimental rats observes that many hepatic enzymes that are inactivated by phosphorylation when the insulin: glucagon ratio is low. Which of the following enzymes are most likely inactivated by this mechanism?

- ✓A. Enzymes 1 and 2
- B. Enzymes 2 and 3
- C. Enzymes 1 and 4
- D. Enzymes 2 and 5
- E. Enzymes 2 and 4
- F. Enzymes 1 and 5

Rationale: Pyruvate kinase (Enzyme 1) is inactivated by cAMP dependent phosphorylation. PDH complex (Enzyme 2) is inactivated when phosphorylated by PDH kinase.



Question #: 14

Thirty men aged 30-35 years are recruited for the study of metabolism in the fasting state. It is observed that they have normal blood glucose levels after 24 hours of fasting. Which of the following pathways is most likely responsible for the

maintenance of plasma glucose levels at this time?

- A. Glycogen → Glucose 6-phosphate
- B. Oxaloacetate → Citrate
- ✓C. Oxaloacetate → Fructose 1, 6-bisphosphate
- D. Pyruvate → Acetyl CoA
- E. Glucose 6-phosphate → Ribulose 5-phosphate

Rationale: The conversion of oxaloacetate to fructose 1,6-bisphosphate represents gluconeogenesis. Liver glycogen stores are depleted by 18-20 hours of fasting. Gluconeogenesis solely maintains the blood glucose levels after the first day of fasting.

Question #: 15

A 15-year-old boy is brought to the physician by his parents because of a 1-week history of unintentional weight loss and polyuria. Laboratory studies show elevated plasma glucose levels and higher-than-normal hemoglobin A1c levels. Urinalysis shows the presence of glucose. Which of the following metabolites is most likely expected to be elevated in this patient's blood and urine?

- A. HMG-CoA
- ✓B. 3-hydroxybutyrate
- C. Methylmalonate
- D. Acylcarnitine derivatives
- E. α -ketoglutarate

Rationale: The vignette describes a patient with type I diabetes mellitus. Ketoacidosis is a complication associated with type I diabetes mellitus.

Question #: 16

A liver biopsy is performed on volunteers aged 25-30 years for a study on liver metabolism. Following the liver biopsy, a sample of the tissue is homogenized and then centrifuged to remove membrane and organelle material, leaving only cytosolic material in the supernatant. Which of the following enzymes is most likely present in the supernatant?

- ✓A. Acetyl CoA carboxylase
- B. Medium chain acyl CoA dehydrogenase
- C. 3-Hydroxy-3-methylglutaryl-CoA lyase
- D. Ornithine transcarbamylase
- E. Carnitine palmitoyl transferase-II
- F. Pyruvate dehydrogenase complex
- G. Fo-F1 ATP synthase
- H. Succinate dehydrogenase

Rationale: The enzymes of fatty acid synthesis and cholesterol synthesis, glycolysis and many enzymes of gluconeogenesis are located in the cytosol. The enzymes of Krebs cycle, ETC, beta oxidation, ketogenesis and urea cycle are located in the mitochondria.

Question #: 17

A 4-month-old girl is brought to the emergency department by her parents because of a sudden onset of seizures. Results of laboratory studies obtained on admission are shown. Which of the following inborn errors of metabolism is most likely in this patient?

- A. von Gierke disease
- ✓B. Pyruvate dehydrogenase deficiency
- C. Classical galactosemia
- D. Glucokinase deficiency
- E. Hereditary fructose intolerance
- F. Glucose 6-phosphate dehydrogenase deficiency
- G. Pyruvate kinase deficiency

Rationale: The vignette describes a very young child with lactic acidosis and normal blood glucose levels. Serum pyruvate levels are also elevated. It is important to be able to distinguish the listed disorders based on clinical features and laboratory findings.

	Patient's values	Reference range
Arterial pH	7.2	7.36-7.44
Blood glucose	88	70-90 mg/dL
Serum lactate	6	0.4-1.7 mmol/L
Serum pyruvate	0.5	0.08-0.16 mmol/L

Question #: 18

A researcher studying metabolism observes that fatty acid de novo synthesis and cholesterol synthesis are active in the liver after a carbohydrate-rich meal. Which of the following enzymes most likely provides the reducing equivalents (NADPH) for both pathways?

- A. HMG-CoA reductase
- B. Acetyl-CoA carboxylase
- ✓C. Glucose 6-phosphate dehydrogenase
- D. Glyceraldehyde 3-phosphate dehydrogenase
- E. HMG-CoA synthase
- F. Pyruvate dehydrogenase complex

Rationale: Reducing equivalents (NADPH) are provided by the pentose phosphate pathway (G6PD) for both fatty acid and cholesterol synthesis.

Question #: 19

A 12-year-old boy is brought to the emergency department by his father after becoming less responsive following several bouts of nausea and vomiting. Laboratory studies show elevated serum acetoacetate and 3-hydroxybutyrate and ketone bodies are present in the urine. Which of the following enzymes is most likely required for the synthesis of these compounds in the liver?

- A. Succinyl CoA: Acetoacetate CoA transferase
- B. HMG CoA reductase
- C. Hormone sensitive lipase
- D. Lecithin cholesterol acyl transferase
- ✓E. Mitochondrial HMG CoA synthase
- F. Acetyl CoA carboxylase

Rationale: The regulated enzyme of ketogenesis is mitochondrial HMG CoA synthase (present only in the liver). The enzyme succinyl CoA:acetoacetate CoA transferase is present in peripheral tissues (not found in liver) and is required for the utilization of the ketone bodies.

Question #: 20

A 20-year-old man comes to the physician because of a 1-day history of passing brown colored urine, muscle pain and muscle cramps after running on the treadmill for about 30 minutes. He says he can tolerate mild exercise. Examination of a muscle biopsy specimen shows excessive glycogen accumulation. Muscle triglyceride levels are within reference range. Examination of a liver biopsy specimen shows normal glycogen structure and content. Which of the following additional laboratory findings will most likely be seen in this patient?

- ✓A. Low blood lactate levels following exercise
- B. Fasting hypoglycemia
- C. Presence of reducing sugar in urine
- D. Decreased muscle carnitine levels
- E. Increased serum C-10 acyl carnitine levels

Rationale: The patient has a glycogen storage disorder affecting the muscle (symptoms and laboratory findings). An important test used in the diagnosis is the measurement of blood lactate levels following exercise. Blood lactate levels increase following ischemic exercise in normal persons due to glycogenolysis and anaerobic glycolysis. When there is a defect in glycogen breakdown (McArdle disease), there is NO increase in blood lactate levels following exercise.

Question #: 21

A 13-month-old girl is brought to the physician by her parents for a follow-up examination. She has a 2-month history of episodes of lethargy that occurs a few hours after the consumption of fruits. Blood glucose levels during these episodes ranged from 45-50 mg/dL (reference range: 70-110 mg/dL). Physical examination shows mild icterus and hepatomegaly. She is advised to avoid a specific

carbohydrate in her diet. Which of the following reactions is most likely impaired in this patient?

- ✓A. Cleavage of fructose 1-phosphate
- B. Formation of UDP-galactose
- C. Phosphorylation of galactose
- D. Phosphorylation of fructose
- E. Substrate level phosphorylation in glycolysis
- F. Intestinal digestion of lactose

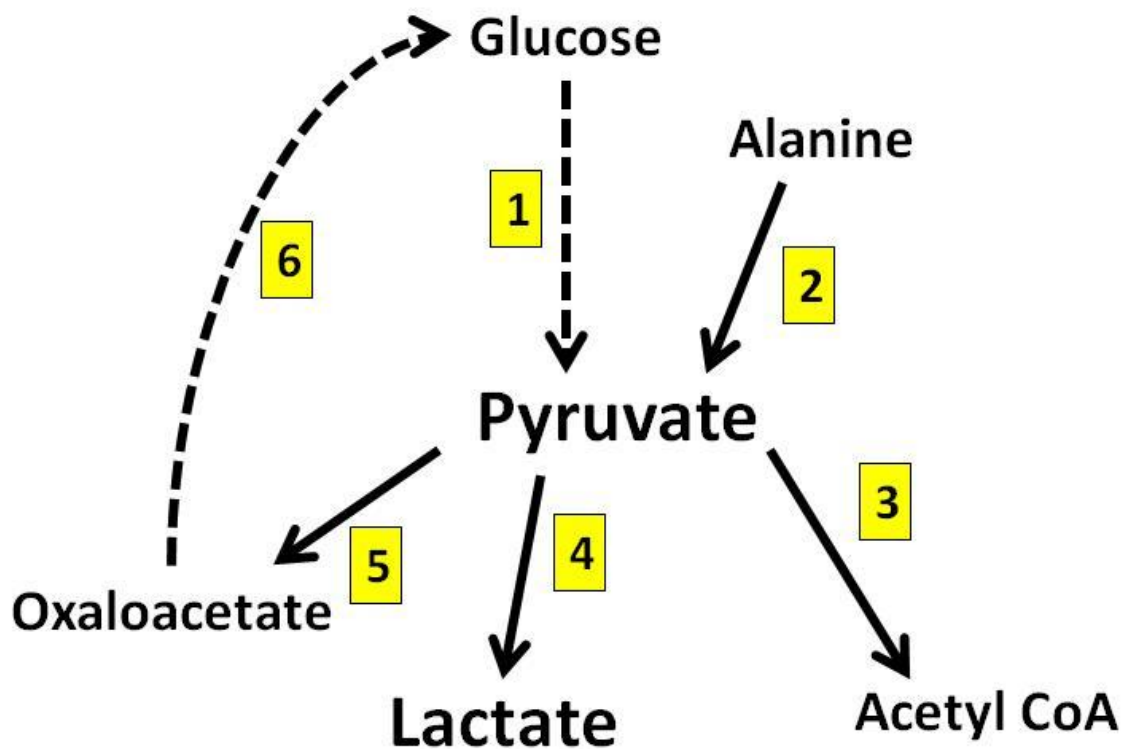
Rationale: The girl in the vignette has hereditary fructose intolerance characterized by hypoglycemic episodes following fruit or sugar consumption and hepatic involvement. The enzyme deficient is aldolase B which is required for the breakdown of fructose 1-phosphate to glyceraldehyde and DHAP. It is important to be able to distinguish and differentiate galactosemia (classical and non-classical), lactose intolerance and benign fructosuria.

Question #: 22

An investigator studying the role of pyruvate in intermediary metabolism identifies various reactions by which it is formed and utilized in various tissues and organs. A flow chart representing its metabolism is shown in the figure. Which of the following statements best describes the pathway shown?

- A. Pathway 1 is inhibited by high concentrations of fructose 2,6-bisphosphate in the liver
- B. Step 5 is inhibited by high levels of acetyl CoA
- ✓C. Step 4 occurs in the actively contracting skeletal muscle
- D. Biotin deficiency results in low activity of Enzyme 4
- E. Pathway 6 is inhibited by glucagon
- F. Enzyme catalyzing step 2 requires TPP as a coenzyme

Rationale: Pathway 1 is glycolysis which is activated by high concentration of fructose 2,6-bisphosphate in the liver. Step 2 is catalyzed by alanine aminotransferase (ALT) which requires PLP as a coenzyme. Step 3 is catalyzed by PDH complex and is a mitochondrial enzyme. Lactate dehydrogenase catalyzes step 4 and uses NADH. There is increased formation of lactate when the skeletal muscle is actively contracting. Step 5 is catalyzed by pyruvate carboxylase which requires biotin as a coenzyme. Pathway 6 is gluconeogenesis which is increased during the fasting state (low insulin glucagon ratio).

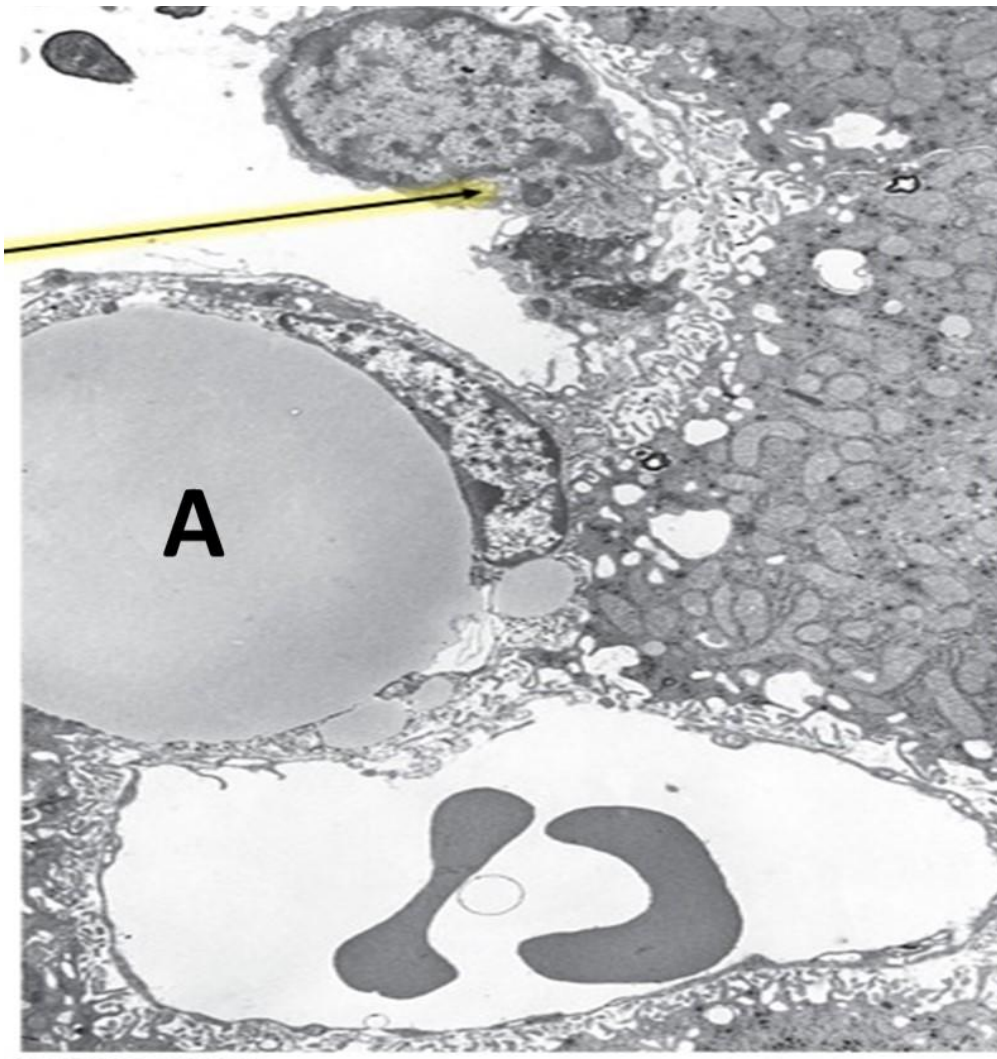


Question #: 23

A 73-year-old man comes to the physician because of a 6-week history of right sided upper abdominal pain and yellowing of the skin. Physical examination shows an enlarged liver with jaundice. An ultrasound-guided biopsy is obtained from the liver and a TEM of a hepatic sinusoid is shown in the attached micrograph. Which of the following is most likely a function of the cell shown at the label **'A'**?

- ✓A. Vitamin A storage
- B. Sense bile composition
- C. Phagocytosis
- D. Bile production
- E. Secretion of nitric oxide

Rationale: The hepatic stellate cells (Ito cells) are found in the perisinusoidal space. These cells are the primary storage site for hepatic vitamin A in the form of retinyl esters within cytoplasmic lipid droplets.



Question #: 24

A 32-year-old man comes to the physician because of a 6-month history of intermittent abdominal cramps and bloating. History shows the symptoms typically appear following a meal rich in carbohydrates. Physical exam shows no abnormalities. Genetic testing shows a mutation in a gene for an enzyme that breaks down oligosaccharides to monosaccharides for absorption. Which of the following best characterizes the localization of this enzyme?

- A. Digestive organelles of hepatocytes
- B. Cytoplasm of enterocytes
- C. Glands of the intestinal submucosa
- D. Lumen of the stomach

✓E. Microvilli of enterocytes

Rationale: Enterocytes produce enzymes that are inserted into the apical plasma membrane that function in the terminal digestion and absorption. They form part of the glycocalyx coating over the apical surface of the enterocyte.

Question #: 25

A 61-year-old man undergoes surgery to remove a peptic ulcer of his stomach. As a result of the surgery, a large number of cells with a basal basophilia usually found at the fundus of the gastric glands are lost. The loss of these cells will most likely cause a decrease in which of the following products?

- ✓A. Pepsinogen
- B. Secretin
- C. Lysozyme
- D. Mucus
- E. Intrinsic factor

Rationale: Chief cells are protein-secreting, basophilic cells located deep in the gastric glands. Chief cells secrete pepsinogen. Parietal cells secrete intrinsic factor, tend to stain eosinophilic and are more concentrated at the neck of the gland. Mucus cells secrete mucus, and are pale staining. APUD cells of the intestine secrete secretin. The APUD cells of the stomach secrete ghrelin. Paneth cells secrete lysosome and are part of the mucosa of the intestine, not the stomach.

Question #: 26

A 51-year-old man is brought to the emergency department because of a 3-hour history of severe abdominal pain. He has been binge drinking alcohol (consumption of excessive amounts of alcohol) over the past 2 days. Physical examination shows acute tenderness to his epigastric region. Laboratory studies of the blood show elevated levels of pancreatic enzymes suggestive of acute pancreatitis. This condition is characterized by autolysis of the secretory units of the parenchyma of the gland. Which of the following hormones stimulate secretion of these enzymes?

- A. Gastrin
- B. Serotonin

- ✓C. Cholecystokinin
- D. Motilin
- E. Pepsin

Rationale: Cholecystokinin (CCK is a polypeptide hormone secreted by the enteroendocrine cells of the duodenum that causes the acinar cells of the exocrine pancreas to secrete their proenzymes.

Question #: 27

A 51-year-old alcoholic man is brought to the emergency department because of a 1-hour history of severe vomiting of bright red blood. Investigation lead to the diagnosis of esophageal varices secondary to portal hypertension. Which of the following is more than likely a feature of the layer of the wall of the esophagus where these varices are usually located?

- A. Stratified squamous epithelium
- B. Myenteric plexus
- ✓C. Mucous glands
- D. Peyers' patches
- E. Smooth muscle bands

Rationale: The submucosa is characterized by large venous plexuses which are tributaries to the portal vein and can become dilated in portal hypertension. In the esophagus, this layer also uniquely contains glands (esophageal glands proper) which produce a slightly acidic mucus.

Question #: 28

A 55-year-old man is brought to the emergency department because of a 1-month history of worsening abdominal pains, constipation and a weight loss of 6.8-kg (15-lb). Rectal examination shows a mass in the distal rectum that bleeds easily. A biopsy of the mass shows cells with multiple mitoses and loss of architecture. Which of the following is most likely a normal histological feature of the structure in which the mass was found?

- A. Well-developed brush border
- B. Intestinal villi
- ✓C. Numerous goblet cells
- D. Submucosal glands

E. Paneth cells

Rationale: While the goblet cells can be present as early as the small intestine, they are most abundant in the large intestine. The rectum is a segment of the large intestine (which is composed of colon, rectum and anus)

Question #: 29

A 45-year-old man comes to the physician because of a 2-month history of headaches, muscle cramps, and fatigue. His blood pressure is 160/110 mm Hg, pulse is 84/min, and temperature is 37°C (98.6°F). Physical examination shows a loud S2 in the right upper sternal border. Laboratory studies show:

Test	Patient value (Reference range)
Serum sodium	145 mEq/L (Normal: 135-145 mEq/L)
Serum potassium	2.6 mEq/L (Normal: 3.5-5.5 mEq/L)
Serum bicarbonate	40 mEq/L (Normal: 22-24 mEq/L)
Plasma aldosterone	80 (Normal: 10-30 ng/dL)

An ECG shows flattened T waves. Ultrasonography of the abdomen shows no abnormalities. Which of the following plasma assays is most likely to confirm the diagnosis in this patient?

- A. Catecholamines
- B. Dehydroepiandrosterone
- ✓C. Renin
- D. ACTH
- E. Cortisol

Rationale: The plasma electrolytes along with the other symptoms of the person indicate that the patient might be having hyperaldosteronism (high plasma Na⁺, low K⁺, and high HCO₃⁻ in addition to hypertension). To differentiate between the primary and secondary hyperaldosteronism plasma concentration of renin would be critical. Plasma renin is generally decreased in primary hyperaldosteronism but increased in secondary hyperaldosteronism. In both hyperaldosteronisms, plasma concentration of aldosterone is increased. Option A is incorrect. Although catecholamines may cause hypertension but not the serum electrolytes imbalances. Option B is incorrect as dehydroepiandrosterone predominantly has androgenic effects and signs and symptoms of this patient cannot be explained rationally by this hormone. Option D and E are also incorrect as these two hormones do not influence the mineralocorticoid

activity.

Question #: 30

A 14-year-old girl is brought to the physician by her mother because of a 10-month history of progressive increase in height, increase in shoe size, and weight gain. The mother says that her daughter is growing taller than her friends. Menarche occurred 3 months ago. She is 6 ft (183 cm) tall and weighs 86 kg (190 lb). Her blood pressure is 140/88 mm Hg; other vital signs are within normal limits. Physical examination shows increased thickness of fingers and toes, coarse skin, and prominent mandible and maxilla. Laboratory studies show elevated random blood glucose and plasma IGF-1. Which of the following is the most likely diagnosis?

- A. Acromegaly
- B. Cushing's disease
- C. Addison's disease
- D. Levi loran disease
- ✓E. Gigantism

Rationale: This patient is around the age of puberty, and therefore, the long bones have not yet fused. During this period, an increase in GH would cause increase in growth of the long bones along with the growth of the muscles, internal organs and other soft tissues. This results in gigantism. Acromegaly is typically diagnosed in adults, after the fusion of the epiphyses of the long bones.